

```

1 import java.util.*;
2
3 class HuffmanNode {
4     char data;
5     int freq;
6     HuffmanNode left, right;
7
8     HuffmanNode(char data, int freq) {
9         this.data = data;
10        this.freq = freq;
11        left = right = null;
12    }
13 }
14
15 public class Main {
16
17     static void printCodes(HuffmanNode root, String code) {
18         if (root == null)
19             return;
20
21         // Leaf node
22         if (root.left == null && root.right == null) {
23             System.out.println(root.data + " : " + code);
24             return;
25         }
26
27         printCodes(root.left, code + "0");
28         printCodes(root.right, code + "1");
29     }
30
31     static void huffmanCode(char[] data, int[] freq) {
32
33         PriorityQueue<HuffmanNode> minHeap =
34             new PriorityQueue<>((a, b) -> a.freq - b.freq);
35
36         // Insert all characters into Min Heap
37         for (int i = 0; i < data.length; i++) {
38             minHeap.add(new HuffmanNode(data[i], freq[i]));
39         }

```

```

41 // Build Huffman Tree
42 while (minHeap.size() > 1) {
43
44     HuffmanNode left = minHeap.poll();
45     HuffmanNode right = minHeap.poll();
46
47     HuffmanNode parent =
48         new HuffmanNode('$', left.freq + right.freq);
49
50     parent.left = left;
51     parent.right = right;
52
53     minHeap.add(parent);
54 }
55
56 // Generate Huffman Codes
57 System.out.println("\nHuffman Codes:");
58 printCodes(minHeap.peek(), "");
59 }
60
61 public static void main(String[] args) {
62
63     Scanner sc = new Scanner(System.in);
64
65     System.out.print("Enter number of characters: ");
66     int n = sc.nextInt();
67
68     char[] data = new char[n];
69     int[] freq = new int[n];
70
71     System.out.println("Enter character and frequency:");
72
73     for (int i = 0; i < n; i++) {
74         data[i] = sc.next().charAt(0);
75         freq[i] = sc.nextInt();
76     }
77
78     huffmanCode(data, freq);
79
80     sc.close();
81 }
82 }

```

```
Enter number of characters: 4
Enter character and frequency:
A 23
B 12
C 34
D 10
```

Huffman Codes:

```
C : 0
D : 100
B : 101
A : 11
```

```
...Program finished with exit code 0
Press ENTER to exit console. 
```